Impact Modelling Report of a Hydrogen Vehicle Chassis

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1. Introduction

In the outlook of an increase in carbon dioxide emissions hydrogen vehicles are an emerging alternative which come to pose a potential solution to the need of studying novel propulsion system solutions and their challenges (Corbo, Migliardini and Veneri, 2011). Therefore, safety measures dictate an area where research should be continued to comply with regulations such as the SAE J2579 (Hosseini and Butler, 2020), for example, to comply with permeation and leakage events which can be assessed by the addition of protective devices such as impact attenuators. Potential is identified in the analysis of high-performance composite materials to implement lightweight solutions to protect the integrity of the fuel cells contained within the analysed chassis enclosure. These solutions should be designed in a way that a progressive crushing event is the result of material failure (Boria *et al.*, 2016).

The outcomes of this investigation will relate mainly to compare and assess different composite impact attenuator designs, implementing a systematic approach to critically choose the best design for the safety cell designed. Afterwards, optimisation is carried out based on dynamic impact simulations which will determine the main drawback of each iteration until the front facing panel is safe of material intrusion. Finally, said final maximum deformation must be below 5 millimetres compared to the baseline design.

Section 2 presents the followed procedure with the relevant parameters and design criteria. In section 3 the results of the baseline geometry versus attenuator assembly are processed. Ultimately, in section 4 a discussion of the results poses the main limitations, drawbacks and highlights to critically assess them.

2. Procedure

2.1 Geometry

The final geometry of the proposed design is shown in Fig. 1.

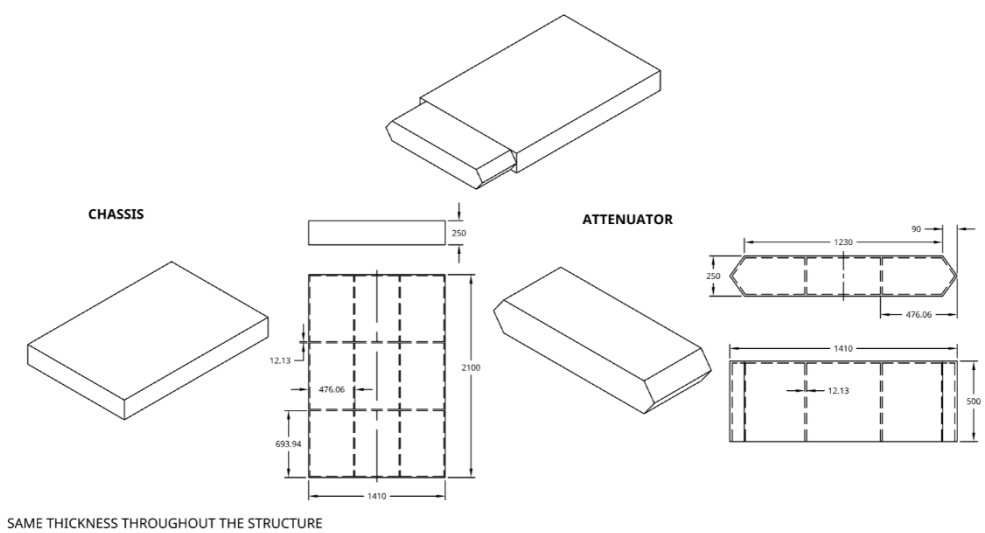

Figure 1. Geometry of the achieved design.

2.2 Meshing procedure

The model was designed in SOLIDWORKS as a 3D solid with the given dimensions in Fig. 1 and then developed in FeMap with a 2D approach extracting the mid surfaces. The surfaces of the chassis were split to align the discretisation when seeding the reinforcements. Both parts in the model are meshed with linear quad shell elements using the Belytschko-Lin-Tsay formulation (Livermore Software Technology, 2022). Finally, the coincident nodes are merged to ensure connectivity when the analysis is done for all the studied parts in the report, including the assessment for the best attenuator. A mesh convergence is performed with support of identical results in preliminary studies to determine the optimal element size (Fig. 2).

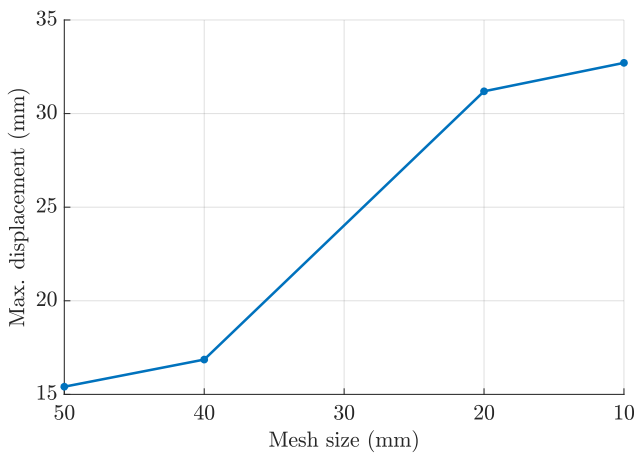

Figure 2. Mesh convergence of maximum displacement at the front panel vs mesh size.

A mesh size of 20 mm is selected to optimise computational resources accounting for the implementation of the attenuator later in the study. This decision balances precision and practicality for the attenuator stage. A total of 23100 four-noded elements are present in the safety cell.

2.3 Baseline analysis set up

The meshed chassis is fixed at the back face base and a primitive block modelled as ‘material type 20 LS-Dyna Rigid’ in FeMap with a mass of 1300 kg impacts it at the front or at the side with a velocity of 13.86 m/s. This is achieved by adding an automatic connection between their property regions. To maintain an even parallel impact, all the other nodal directions in the impactor are restricted. More analysis details can be referenced in Table 1.

Table 1. Parameters used in the setup of the analysis.

Analysis section	Item	Value
Connections	Static friction coefficient	0.3
	Dynamic friction coefficient	0.25
Set up	Analysis type	Explicit Transient Dynamics
	Output time interval	0.001 s
	Cores	4

As a greater deflection was identified at the front impact, this scenario is considered for the further development of the analysis. More detail on these values can be found in the results section.

2.4 Attenuator stage assessment

As explained before, the attenuators are processed using 2D surfaces in FeMap which enables to reduce computation time and analyse crushing mechanisms at the cost of not representing delamination phenomena (Oñate *et al.*, no date; Oshinibosi and Barton, 2012), which in this case is not a priority. Therefore, the assessment process and base shapes for the best design are presented in Fig. 3a and the developed iterations in Fig. 3b.

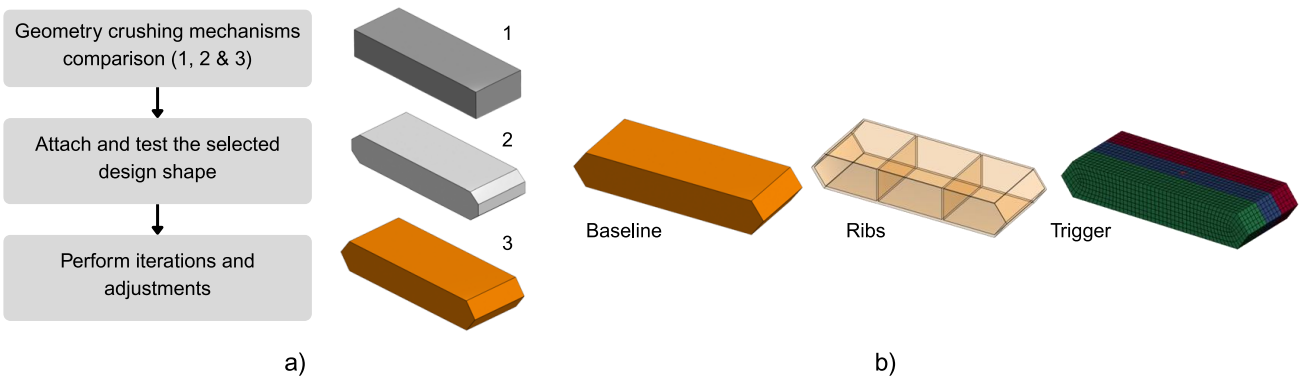

Figure 3. a) Workflow for the assessment of the best attenuator; b) Proposed attenuator iterations.

The nodes at the base of the attenuators are fixed in all degrees of freedom (Boria *et al.*, 2016) to analyse their crushing behaviour prior to adding a defined design to the chassis. Similarly, these nodes are then tied to the elements of the front panel of the chassis and an automatic contact is added to impact both models. Finally, a general one-way contact is applied to the impactor to account for folding contacts. The same velocity conditions are maintained, and the number of quad elements range from 7800 to 8500 depending on the design with a size of 15 mm for the preliminary comparison.

2.5 Material properties and formulation

The material orientation is aligned parallel with the impact direction and different lay ups are used for the chassis, reinforcements and attenuator respectively (Fig. 4). A greater thickness, and consequently more plies, are used at the inside region of the safety cell whilst the attenuator is kept at a smaller thickness to promote crushing. More material details are found in Table 2.

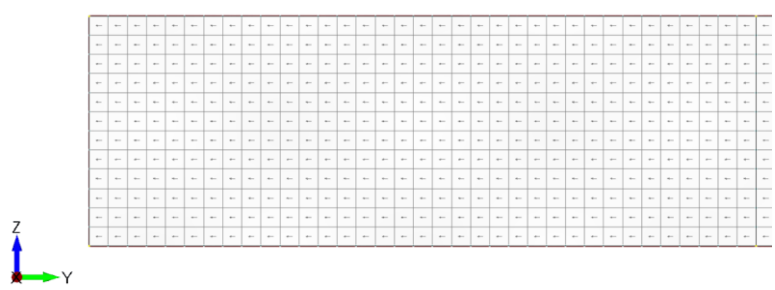

Figure 4. CFRP material orientation for the chassis.

Table 2. Main properties and reference information for the materials used.

Section	Material used	Main properties	Reference
Chassis envelope	M55 UD	$E_{11} = 300\ GPa$	(Performance Composites, no date)
		$G_{12} = 5\ GPa$	
	Easy Cell 75 Foam	$\rho = 75\ kg/m^3$	(Easy Composites, no date)
		$G = 27.75\ MPa$	
Reinforcements	150 UD	$E_{11} = 223\ GPa$	(Gurit, no date)
		$G_{12} = 4.31\ GPa$	
Attenuator	M55 UD	$E_{11} = 300\ GPa$	(Performance Composites, no date)
		$G_{12} = 5\ GPa$	

The model of the composite materials is number 54, which is suitable for unidirectional materials (Fellows, 2017). Additionally, material model number 38 is selected to include a foam layer in the layup of the chassis suitable for shell elements (Ansys, no date), with 8 outer laminas being oriented at ± 45° and 6 inner longitudinally with the direction of impact. This results in a total thickness of 12.13 mm (0.152 mm p/ply). The reinforcements are modelled with a total thickness of 16 mm using the same ply thickness as the chassis. Separately, the final design of the attenuator integrates a trigger mechanism (Oñate *et al.*, no date) which is a progressive thickness variation, a layup representation can be found in Fig. 5.

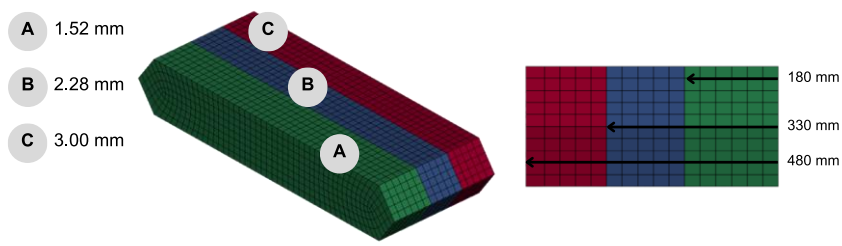

Figure 5. Schematic representation of the proposed thickness variation in the final attenuator.

3. Results

As mentioned in section 2.3, the front impact represents in this case the worst scenario which will be analysed throughout the report. In Fig. 6 a comparison is made between 2 representative nodes at the front and side panel for a baseline impact of the chassis enclosure. The data is obtained from FeMap and processed in MATLAB

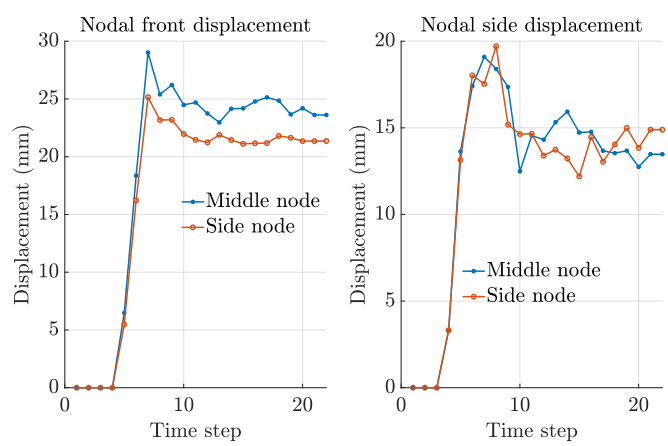

Figure 6. Nodal displacement comparison between front and side impact with two measured nodes.

to obtain the displacement versus time plots, a common procedure throughout the presented graphs.

It can be noticed that in the front scenario a greater deflection, by 10 additional millimetres, is estimated at the central region of the facing panel. In addition, computational resources are managed by setting up a minimum threshold time 'TFAIL' of 1.5E-8 s, this will limit and delete highly distorted elements which do not contribute on the results anymore (Feraboli *et al.*, 2011).

A validation is done in Fig. 7 by estimating an expected maximum deflection at 4 m/s with hand calculations derived from a conducted laminate analysis compared to a subsequent simulation. The hand calculations overestimate the experimental results by 0.7 mm which show an acceptable level of agreement and acceptance for further deflection values obtained.

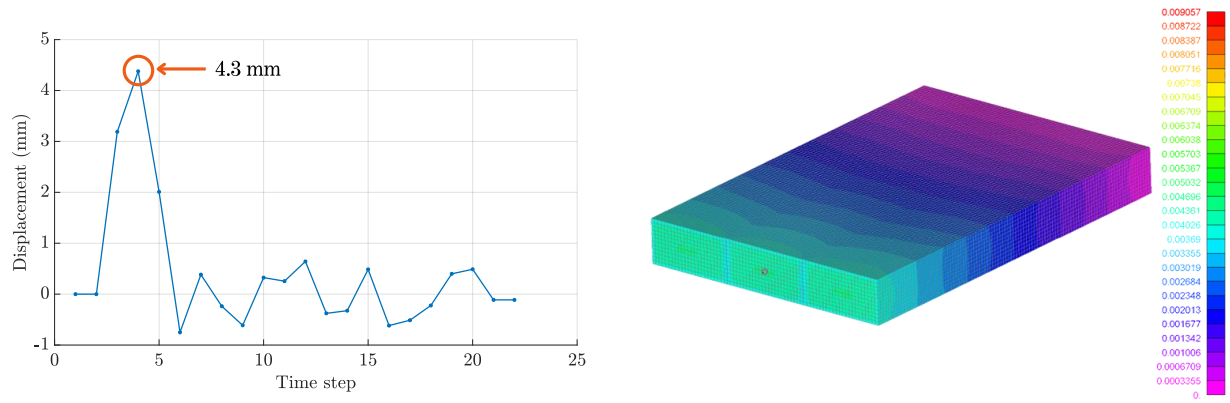

Figure 7. Baseline validation run.

Fig. 8 shows the deceleration curves of the different attenuator shapes. It is identified that the hexagonal shape has the best performance with a more gradual load transfer, progressive deceleration (oscillations in the graph) and lower peak forces. However, a compromise is made in terms of the Specific Energy Absorption (SEA) obtained integrating numerically. Thus, priority is given to the folding characteristics for the latter addition to the chassis.

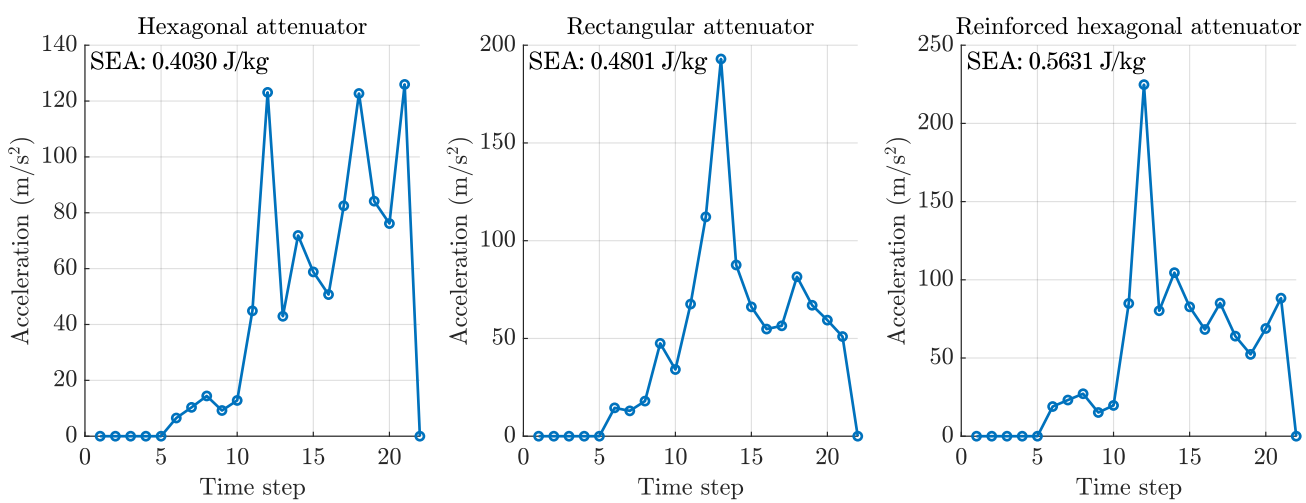

Figure 8. Deceleration curves for the three tested configurations.

The resulting deformations throughout the iterations conducted for the selected shape are shown in Fig. 9. To visualise the scenarios, the specific impact time step is set in Ls-PrePost isolating the front panel and attenuator regions. The best solution is achieved with structure ribs and a progressive thickness.

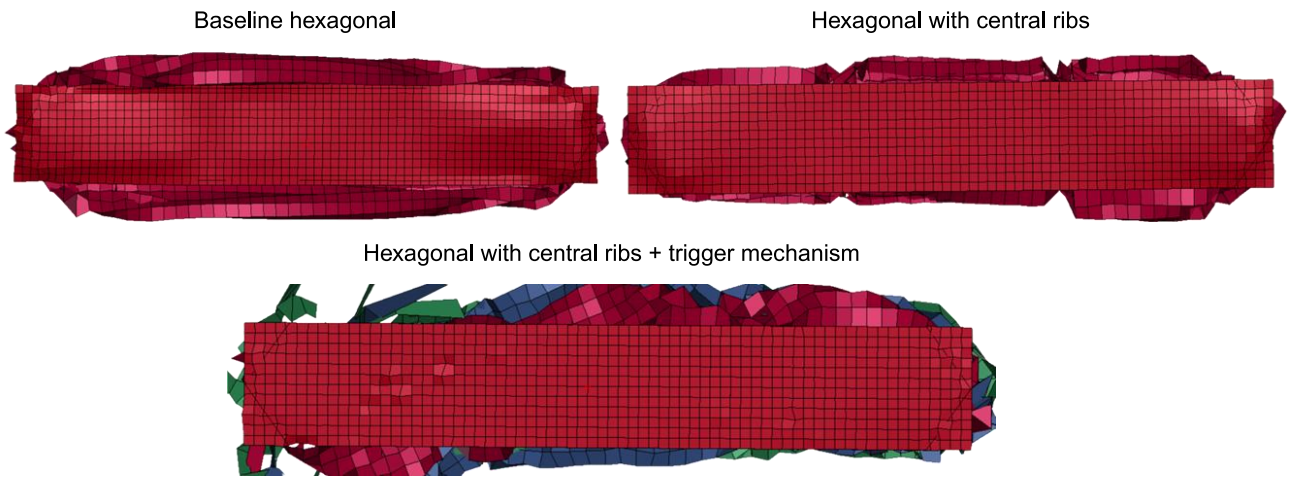

Figure 9. Deformation assessment of the proposed iterations to improve the hexagonal attenuator.

To have a direct comparison, the resultant deformations on the axis of the impact are plotted and compared in Fig. 10. The data for the attenuated assembly is obtained from Ls-PrePost and processed in MATLAB based on the d3plots files generated for the simulation.

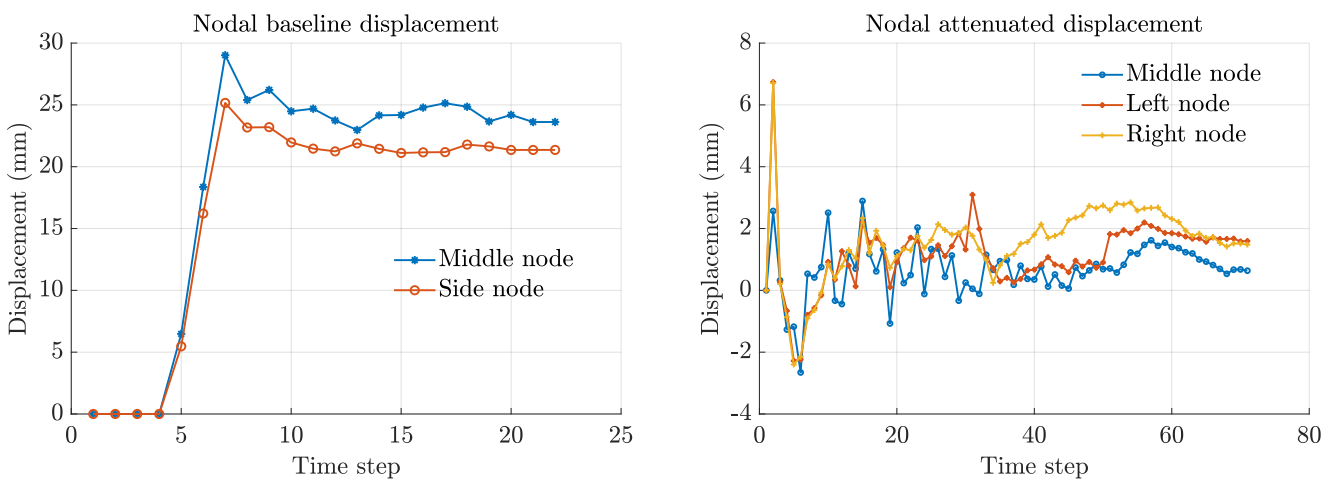

Figure 10. Displacement comparison between baseline design and proposed attenuated assembly.

Finally, the acceleration and kinetic energy of the impactor body is shown in Fig. 11. The initial peak in the change of velocity correlates with the identified behaviour of the displacement after installing the attenuator. Meanwhile, the safety device achieves a reduction of all the kinetic energy of the moving wall.

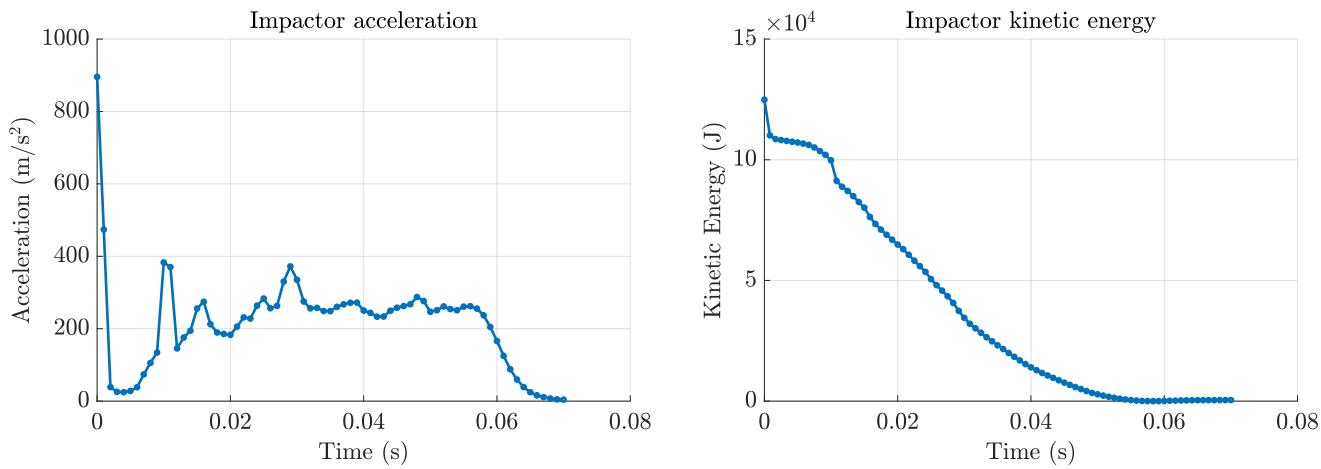

Figure 11. Acceleration and kinetic energy of the moving wall.

4. Discussion

Throughout the study some computational inconsistencies were found at recurrent time step variations between simulations, this was accompanied by an increased demand in computational resources which was verified by analysing identical runs on Windows operating computers with 32 and 64 GB respectively. Based on this, the root cause is the calculation of the compression of the crushed carbon fibre of the attenuator, which made the subsequent models finish from 10 minutes in baseline configuration to up to 120 minutes using 4 cores. Evident material failure changes between the first two attenuator iterations are clear compared to the last trigger design in Fig. 10 presenting ambiguity. Although there was a lack of consistency in the impact deformation at the last instance the qualitative results show good agreement with pertinent solutions to implement when trying to reduce intrusion into a safety cell with hydrogen cells, which is the most important outcome of this study, to achieve a reduction in an already stiff structure from 29 mm to 6 mm at the initial peak displacement, afterwards maintaining minimum oscillations below 5 mm.

The progressive attenuator design is deemed operationally successful in reducing completely the kinetic energy from the impactor with controlled acceleration oscillations at the front panel after the initial transient has come to a steadier response. A sensible systematic approach returned an expected trend where priorities in lower peak forces and progressive folding events in the proposed attenuator are reflected in the acceleration of the rigid body. However, it must be stated that there are numerical instabilities at the later to end stage of the impact, causing inner attenuator elements to propagate a failure in the front panel of the chassis. Thus, the acceleration response gives clear conclusions regarding the performance of the trigger folding in the attenuator, as there is a significant deceleration at the beginning but then the impactor, with enough kinetic energy keeps accelerating for a fraction of time before it is abruptly decelerated by the next trigger region. This is repeated until the crushed material effectively attenuates the remaining momentum of the moving wall.

A solution proposal for the previous computational challenges was investigated by using a coarser mesh in the attenuator, this did improve initial processing managing resources but at late stages of the analysis did not make a difference. Adding hourglass viscosity effects (Fellows, 2017) and conducting a complementary analysis with changes in material properties to assess these variations is proposed a future work. Similarly, the thickness in the last progressive design can be thoroughly optimised according to the length to width ratio of the front face of the chassis, as this will bring conclusive results regarding initial numerical inconsistencies or needed design improvements. Another alternative can be presented in a globally coarser mesh with local refinements at interest regions (e.g. front panel or attenuator).

5. References

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